
When Shift Happens - Confounding Is to Blame

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Abstract

Across machine learning applications, distribution shifts create uncertainty about whether patterns learned in one environment will remain useful in another. Although a common route to robustness is to learn causal-invariant representations, recent empirical studies point to a broader operating regime: (i) empirical risk minimization (ERM) can match or exceed state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) OOD performance can improve when models use all available covariates rather than only variables judged to be causal. We explain this behavior through hidden confounding. When the distribution of an unobserved confounder changes across environments, the observed data can exhibit several shifts at once, invalidating assumptions used by many OOD objectives. In this setting, we prove that generalization may require learning environment-specific relationships in addition to invariant ones. We further show that augmenting models with observed proxies for hidden confounders can reduce the difficulty of hidden confounding shifts. Together, the theoretical and empirical results broaden guidance for OOD learning and covariate selection in settings where unobserved factors plausibly shape both covariates and outcomes.

1 Introduction

Generalization—drawing reliable conclusions about unseen data from observed data—is a shared requirement across scientific and engineering domains. In medicine and the social sciences it appears as *external validity*, where findings from one population should transfer to another [13]; in ecology it motivates *space-for-time substitutions*, where spatial variation is used to reason about long-term temporal change [60]; and in engineering it underlies *robust control*, where systems must perform despite unmodeled disturbances [35]. Machine learning now contributes to this broader goal by fitting predictors intended to remain useful beyond the exact conditions under which their training data were collected [100, 88, 49].

For learned predictors, this requirement is commonly formalized as robustness to distribution shifts between training and deployment data. A model trained to predict *food stamp reciprocity* from *household attributes* in one state, for example, may be deployed in another state whose population and policy context differ. We say a model achieves *out-of-distribution (OOD) generalization* when it preserves useful performance on both in-distribution (ID) data, from which training samples are drawn, and OOD test data. Domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94] offer objectives with guarantees under specific assumptions. Yet, with careful model selection [30, 38, 87], standard *empirical risk minimization (ERM)* [86] often remains competitive across real-world OOD benchmarks [30, 24, 50, 57, 67, 87]. A recent study of 16 real-world tabular datasets further found that using all observed covariates, including variables without direct causal effects on the outcome, can improve OOD performance [57]. These observations suggest that practical generalization can depend on a wider set of statistical signals than causal-invariant features alone.

Distribution shifts often arise because data are collected from multiple environments [71]. Food stamp reciprocity can vary by state, for instance, because eligibility rules and population composition differ, producing environment-specific distributions. Some covariate–outcome relationships may nevertheless remain stable across environments; these are commonly called *invariant relationships* [6].

Many OOD methods seek such relationships [6, 59, 31, 66, 56, 98, 63]. A particularly important case is *causal invariance*, where causal mechanisms are stable across environments [71, 59, 6]. In many applied datasets, however, relevant variables are unobserved because of collection limits, privacy constraints, or measurement error [14, 52, 20, 48]. When such variables confound both observed covariates and outcomes, changes in their distributions can make the observed invariant structure disappear. Treating these hidden confounding shifts as ordinary covariate or label shifts can therefore reduce OOD performance [40, 4, 85, 62] and can cause models to learn misleading observed associations.

Although hidden confounding is common in applied data analysis, its consequences for OOD generalization have received comparatively limited attention [4, 40, 85, 62]. This work develops a causal and information-theoretic account that helps explain empirical patterns reported across recent OOD studies [30, 24, 50, 57, 67, 87]. Our scope is not to introduce new datasets or training objectives, but to clarify when existing objectives and covariate choices can succeed or fail under hidden confounding shift. Our contributions are as follows:

- We motivate confounding shift as a broadly relevant causal source of OOD difficulty (§ 3) and explain why informative, non-causal covariates can be useful rather than merely spurious.
- We show that maximizing predictive information between predictions and outcomes can require environment-specific relationships; invariant representations remain important, but are not by themselves sufficient in this setting (§ 4.1, § 4.2).
- We prove that variables informative of either the outcome or hidden confounders can improve predictive information, giving a general rationale for covariate selection when hidden confounding shift is plausible (§ 4.3).
- Through real-world and synthetic datasets, we provide evidence that (i) hidden confounding is visible in tabular benchmark data, (ii) learning environment-specific relationships correlates positively with ID and OOD test accuracy, and (iii) informative, non-causal covariates can improve generalization under the evaluated conditions (§ 5).

Overall, the paper reframes the effectiveness of ERM and all-covariate models as a consequence of hidden confounding structure that can appear across many OOD tabular prediction settings, while also identifying the conditions under which methods must go beyond invariance.

2 Related work

OOD generalization sits within a broader family of approaches for learning under changing data-generating conditions, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These settings differ in what information is available: domain adaptation assumes unlabeled target-domain data, domain generalization uses environment labels during training, federated learning handles distributed multi-environment data under communication and privacy constraints, and OOD detection identifies samples that depart from the training distribution. Across these settings, invariance is a central organizing principle, but recent work argues that invariance alone does not fully characterize generalization [46, 72, 93]. In particular, Ye et al. [93] emphasize that feature informativeness must complement invariance. We study the same invariance–informativeness tradeoff in the specific but broadly occurring case of hidden confounding shift.

Proxy variable adjustment: When confounders are available during training but unavailable at test time, Landeiro and Culotta [40] adjust for confounding shifts to improve classification. Proxy-based causal effect identification [55, 39] has also inspired domain adaptation methods that treat unknown distribution shift as an identification problem [4, 85]. More recent OOD work studies hidden confounding shift under overlapping confounder support [62]. Our contribution is complementary: rather than relying on a single proxy-adjustment recipe, we characterize why proxy variables can help OOD prediction by changing the information terms that govern generalization.

All variable models vs causal models. Recently, Nastl and Hardt [57] introduced a benchmark study where covariates are categorized into four groups: causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates. They show that across 16 benchmark datasets, models using all covariates Pareto-dominate those using only causal or arguably causal subsets on both ID and OOD data. However, there is limited theoretical work explaining these results. We present scenarios and arguments to explain their experimental findings. For linear causal models, anchor

regression [68] introduces a framework that balances between two estimation paradigms: models that include all observed covariates and models that focus solely on causal covariates. We aim to explain the impact of adding more covariates that are not necessarily causal under a hidden confounding shift. Eastwood et al. [21] show that unstable covariates can boost performance when they carry information about the label, provided they are conditionally independent of the stable covariates given the label. They propose to adjust the distribution shift by looking at the test domain without labels. However, when applied to a medical real-world dataset not constructed for this particular problem [8], ERM still remains competitive with their method, in line with the findings of Nastl and Hardt [57]. This reflects the broader insight that, under well-specified covariate shifts, maximum likelihood estimation (MLE) achieves minimax optimality for OOD generalization [26]. Our focus is the less idealized regime in which hidden confounding shift makes those well-specified assumptions unreliable in otherwise standard real-world prediction tasks.

3 Manifestations of hidden confounding shift

This section places hidden confounding shift within the broader taxonomy of distribution shifts and uses causal effect identification to illustrate why the same issue can affect prediction across environments.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , environment changes can appear as several observed shifts between e and e' , as summarized in Table 1. A unifying source of these shifts is a change in the distribution $\mathbb{P}(U)$ of an unobserved variable U that causes $\mathbf{X}$, Y , or both (Figure 1). Although U is not observed, changes in $\mathbb{P}(U)$ propagate to distributions over the measured variables. For example, $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, so $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ together with $U \rightarrow \mathbf{X}$ implies $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same reasoning extends to shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

Table 1: Common distribution shifts induced by environment changes in observed variables.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

Many OOD approaches can be interpreted as assuming causal relationships among U , $\mathbf{X}$, Y that make particular conditional distributions invariant. For instance, when $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$, a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, but conditional covariate distribution stays invariant i.e., $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to covariate shift, i.e., $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, but the conditional distribution $\mathbb{P}(Y | \mathbf{X})$ stays invariant i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. Those invariances need not hold when U causes both $\mathbf{X}$ and Y (Figure 1 (c)), a structure that can occur in many applied datasets [50, 4, 40, 85, 62, 65, 64]. Then a shift in $\mathbb{P}(U)$ can simultaneously induce label shift, covariate shift, conditional covariate shift, and concept shift. The causal system may still satisfy an invariance, $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, known as *stable confounding shift* [85, 4]; however, because U is hidden, this stabilizing relationship is not directly available to the learner.

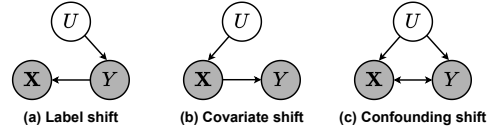


Figure 1: Causal graph patterns that can give rise to different observed distribution shifts.

Information-theoretic measures: Following Federici et al. [22], we use mutual information to quantify distribution shifts. Specifically, we use $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E | Y)$, $I(Y; E | \mathbf{X})$, $I(\mathbf{X}, Y; E)$ where E is the environment variable to measure the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ respectively across the environments. As discussed earlier, existing methods for OOD generalization often assume either $I(\mathbf{X}; E | Y) = 0$ through the label shift assumption or $I(Y; E | \mathbf{X}) = 0$ through the covariate shift assumption. However, as discussed earlier, both of them may be nonzero under hidden confounding shifts.

Challenges with hidden confounding shift: To illustrate the challenges posed by hidden confounding shift on OOD generalization, we present the experiment shown in Figure 2. The outcome Y depends on X and a hidden confounder U . Environment-specific shifts in the mean of U , denoted by μ_e ,

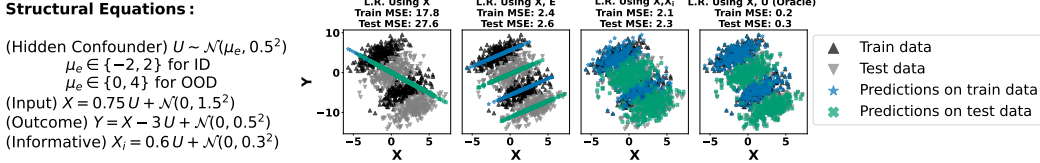


Figure 2: Linear regression (L.R.) models under hidden confounding shifts with limited environment overlap (i.e., distant μ_e). (i) Training only on X can learn the wrong observed association with Y , illustrating Simpson’s paradox. (ii) Environment-specific summaries of X , denoted by E , recover the correct relationship but have limited representation power. (iii) An informative covariate X_i that proxies U improves OOD generalization. (iv) The oracle model observes both X and U .

induce distinct environments. A linear regression model trained solely on observed covariate X infers an incorrect relationship between X and Y , as shown in the first subplot. This exemplifies the Simpson’s paradox [73], where Y increases with X within each environment, yet the model captures the opposite trend. We observe similar behavior from the models designed for OOD generalization, such as IRM [6], VREX [38], and Group DRO [69]. In such scenarios, we theoretically show in § 4 that the optimal strategy for generalization involves learning environment-specific relationships, which enables recovery of the correct relationship between X and Y . This is illustrated in the second subplot, where environment-specific summary statistics of X , such as mean, standard deviation, and quantiles, help uncover the true X – Y relationship. This mirrors the *backdoor adjustment* criteria in causal effect estimation [58], where environment-specific information acts as a proxy for adjusting for confounders. However, as shown in the second subplot, environment-specific information encoded in observed covariates alone may be insufficient due to the limited representational capacity. However, as we show in § 4.3, additional informative covariates serving as proxies for unobserved confounders can further improve generalization performance, as shown in the third subplot. For comparison, we include an oracle model trained on X and U , which achieves the best-possible mean squared error.

4 Impact of hidden confounding shifts on OOD generalization

This section analyzes how hidden confounding shifts affect several general components of OOD prediction objectives. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X, Y}[\log \frac{\mathbb{P}(X, Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The goal in OOD generalization is to learn a predictor $(f \circ \phi)$ that performs well on both ID and OOD data. To this end, various objectives have been considered in the literature. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments: $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ where $\mathcal{E}_{tr}$ denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ where β is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ with the constraint that f is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) based methods simply pool data from all training environments and minimize the empirical risk on the pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$ where $\mathcal{D}^e$ denotes observed data in an environment e .

To connect hidden confounding shift with common objectives, we study predictive information, $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity measures how much the prediction reveals about the true label and therefore abstracts across classification and regression-style prediction goals. We begin with two representation properties, *informativeness* and *invariance*, that are widely relevant for OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness captures how much information $\phi(\mathbf{X})$ carries about Y after fixing the environment. Low CINF corresponds to lost label-relevant information, whereas high CINF can be

problematic in applications such as algorithmic fairness if the representation uses sensitive or biased within-environment signals. Reducing $I(\phi(\mathbf{X}); E)$ can suppress such signals, but it may also remove predictive information under hidden confounding shift, as shown in § 4.2. In the setting analyzed here, where E is not treated as sensitive, we examine the regime in which increasing CINP supports generalization, and we justify that perspective in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$*

Invariance measures whether the representation $\phi(\mathbf{X})$ remains stable across environments after conditioning on a label. Too little invariance can preserve environment-specific structure and encourage overfitting, whereas greater invariance removes dependencies that many OOD methods seek to avoid. Building on the shift measures in § 3, we use $I(\phi(\mathbf{X}); E)$ for *feature shift* and $I(Y; E | \phi(\mathbf{X}))$ for *concept shift*. Depending on whether we discuss raw covariates or learned representations, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

To make the decomposition applicable beyond a single example, we begin with causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are marginally independent but become conditionally dependent when conditioning on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$. If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$. A path is open if it is not blocked.

Consider predictive information $I(Y; \hat{Y})$, where predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are formed from a learned representation $\phi(\mathbf{X})$. Figure 3 gives the causal structure used for the decomposition over $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$, and E , with U representing a hidden confounder. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$, $\hat{Y}$ is redundant for reasoning about Y , so $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. The environment variable E encodes shifts in $\mathbb{P}(U)$ across environments. We first give a general predictive-information decomposition without specifying how U influences $\mathbf{X}, Y$; § 4.2 then specializes the analysis to that influence.

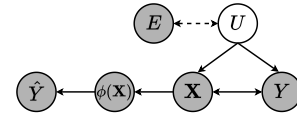


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y represents the general case in which some covariates in $\mathbf{X}$ can cause Y , while others may be caused by Y .

Proposition 4.1. *For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Appendix § A contains the proofs. The decomposition also gives a common lens on existing methods, which can be interpreted as minimizing or maximizing particular terms in Equation (1). For instance, IRM [6] enforces that a fixed classifier f remains the same across different environments. This is directly related to maximizing $-I(\phi(\mathbf{X}); E|Y)$ because if $\phi(\mathbf{X})$

were to carry extra environment-specific information conditional on Y , the fixed classifier would no longer be optimal in all environments. DANN [23] and the independence criterion for fair classification [22] aims to minimize environment-specific information in $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$. In the context of domain adaptation, certain properties of features are learned invariant to domains [82, 90]. While minimizing $I(\phi(\mathbf{X}); E)$ may be good for fair classification, it may degrade predictive performance [98, 33, 22]. This can be seen from Equation (1), where minimizing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to obtain $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. By enforcing this equality, E becomes independent of the pair $(\phi(\mathbf{X}), Y)$. Consequently, $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, although CDAN does not explicitly include $I(Y; E | \phi(\mathbf{X}))$ in its loss, its joint-distribution alignment objective effectively drives this conditional mutual information to zero. While $I(Y; E)$ is a constant, there exist methods that are robust to shifts in the distribution of labels [69, 95].

Equation (1) deliberately abstracts away the specific causal role of U . Once that role is specified, the same terms in (1) yield sharper guidance for OOD learning. That is, if $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, suggesting that minimizing $I(\phi(\mathbf{X}); E | Y)$ is a principled objective. Similarly, if $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, thereby motivating the minimization of $I(Y; E | \phi(\mathbf{X}))$. However, if $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, the interaction between the terms in Equation (1) becomes non-trivial, and it remains unclear what constitutes an ideal strategy for addressing them. We answer this question in the next section.

4.2 Predictive information under hidden confounding shift

To characterize how a hidden confounder U that causes both $\mathbf{X}$ and Y affects predictive information, we analyze *two representative cases* for the observed relationship: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, we obtain the inequalities in (2) because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y which results in additional information flow from $\phi(\mathbf{X})$ to E via Y . In contrast, conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Similarly, when $Y \rightarrow \mathbf{X}$, we obtain the inequalities in (3) because conditioning on Y blocks the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ because $\phi(\mathbf{X})$ is the child of $\mathbf{X}$ and conditioning on the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

Proposition 4.2. *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

Proposition 4.2 treats $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately. Because $\mathbf{X} \rightarrow Y$ is common in real-world tabular prediction tasks. For instance, 11 out of 16 datasets considered in recent benchmark studies [57, 50, 24] follow the causal structure $\mathbf{X} \rightarrow Y$. In the remaining 5 out of 16 datasets, the number of covariates that are caused by Y is much smaller than the number of covariates that cause Y . We present the key takeaway from Proposition 4.2 below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$, (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

284 **Maximizing conditional informativeness:** In a very recent work, Prashant et al. [62] proposed a
 285 mixture of experts (MoE) model for OOD generalization under hidden confounding shift, where
 286 each expert corresponds to a specific hidden confounder assignment. That is, each expert focuses on
 287 maximizing the performance within the environment corresponding to a particular value of the hidden
 288 confounder. Thus, learning MoE models aligns with the goal of maximizing the conditional infor-
 289 mativeness $I(\phi(\mathbf{X}); Y | E)$. Equation (4) provides theoretical justification for the use of MoE-type
 290 models. Beyond supporting such models, the Equation (4) also highlights the need for methods that
 291 operate under more general confounding shift settings. For instance, the method proposed by Prashant
 292 et al. [62] assumes overlapping confounding support i.e., $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, and
 293 the proxy variables to be discrete-valued.

Figure 2 shows that even in a simple setting with no confounder overlap and a continuous-valued proxy, a linear regression model can recover the correct relationship between observed covariates and the target when it receives environment-specific information. This example motivates OOD methods that remain effective when confounder support is limited and the hidden confounder is continuous-valued.

294

295 **Minimizing residual:** The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ captures label-relevant information in $\phi(\mathbf{X})$
 296 that remains unused by the prediction $\hat{Y}$. The residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) =$
 297 $H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ is related to the expected cross-
 298 entropy loss as: $\mathbb{E}_{Y, \hat{Y}}[\ell_{CE}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$
 299 is known as the *refinement error* that measures the model’s ability to distinguish between classes.
 300 Minimizing the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus involves two complementary strategies. First, one
 301 can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making $\phi(\mathbf{X})$ conditionally independent of the target Y
 302 given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can minimize the refinement error $H(Y | \hat{Y})$, for
 303 instance, through training techniques that minimize refinement error for accurate predictions before
 304 minimizing the calibration error [10].

305 4.3 Impact of additional informative variables

306 In § 4.2, the analysis identifies useful generalization strategies when the confounder is unobserved.
 307 Many practical datasets nevertheless include proxy variables for the hidden confounder [4, 85, 62].
 308 We therefore study how adding non-causal but informative covariates changes predictive information.

309

310 **Definition 4.3** (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,*
 311 *neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are*
 312 *not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

313 Because $U \rightarrow Y$, covariates informative about U are also informative about Y , and conversely in this
 314 setting. The decomposition in (1) shows that maximizing $I(Y; \hat{Y})$ requires reducing concept shift
 315 $I(Y; E | \phi(\mathbf{X}))$. Empirically, Liu et al. [50] suggest that collecting additional covariates $\mathbf{X}_I$ that
 316 make $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ more stable across environments can improve OOD test accuracy. We provide
 317 a theoretical account of this broader pattern by showing how informative variables move the terms
 318 in (1) in beneficial directions.

319 **Proposition 4.3.** *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that*
 320 *$Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) &> I(\phi_1(\mathbf{X}); Y | E) & (ii) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) &> I(\phi_1(\mathbf{X}); E) \\ (iii) \ I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) &< I(Y; E | \phi_1(\mathbf{X})) & (iv) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) &> I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

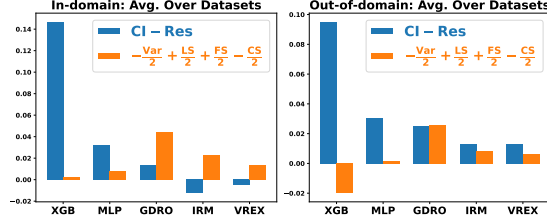
321 Proposition 4.3 shows that informative covariates increase conditional informativeness (5.i) and
 322 feature shift (5.ii), while reducing concept shift (5.iii). The same addition can also increase variation
 323 (5.iv), so broader covariate use may need to be paired with strategies that control environment-specific
 324 variation.

325 5 Experimental results

326 We use real-world and synthetic datasets to evaluate the paper’s claims under the same benchmark
 327 conditions throughout: (i) whether hidden confounding shift is detectable in real-world data, (ii) how

Table 2: Distribution-shift measurements across the evaluated real-world datasets. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. One-sample t-tests against a mean of zero ($\alpha = 0.05$) indicate that all measures differ significantly from zero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Across the eight evaluated datasets, the plotted difference *conditional informativeness* – *residual* is positively correlated with the average ID and OOD test accuracy summarized in the table.

the terms in (1) relate to OOD generalization under hidden confounding shifts, and (iii) whether informative covariates improve generalization. The real-world experiments cover eight tabular benchmark datasets, *Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*, with domain splits from the *TableShift* benchmark [24]. We report ID and OOD test accuracies for two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain generalization methods, IRM [6] and VREX [38]; and one robust learning method, Group DRO (GDRO) [69]. Unless otherwise stated, results average over 5 random hyperparameter choices described in Appendix B; different random seeds have statistically insignificant effects for these datasets [24]. Appendix § B gives the full setup, and Appendix C reports hyperparameter-tuned results.

Hidden confounding shift: Hidden confounding shifts can manifest through observable changes in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. Table 2 shows that label shift, covariate shift, conditional covariate shift, and concept shift all appear in the evaluated real-world datasets, which is consistent with hidden confounding shift rather than a single isolated shift type. As a qualitative complement, we query GPT-4o [1] for plausible hidden confounders in several benchmark datasets [24, 57]; Appendix § E reports these results. In *food stamps*, for example, state-specific *economic policies* may influence both *household income* and *food stamp reciprocity*.

Conditional informativeness vs. accuracy: From Proposition 4.2, under hidden confounding shift, maximizing the difference *conditional informativeness* – *residual* is essential for maximizing predictive information. Results on the first six datasets (that have more than one training environments [57, 24]) listed in Table 2 are shown in Figure 4. Dataset-wise results are in Appendix C. These results show that the difference *conditional informativeness* – *residual* is positively correlated with ID and OOD accuracy, with XGB being the best performing model with highest value of *conditional informativeness* – *residual*. As shown in the dataset-wise results in Appendix C, we observe that in many settings, due to hidden confounding shifts, the bars denoted with orange are close to zero because the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ is zero under hidden confounding shift (Proposition 4.2). The relative ID / OOD performance of the methods shown in Figure 1 is consistent with previously reported results in the literature [57, 24].

Informative covariates vs. accuracy: We next examine whether adding covariates that may be informative, even when not strictly causal, improves OOD generalization in the benchmark setting. To this end, we adopt the covariate partitioning from [57], where all available covariates are grouped into three nested covariate subsets: *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates*

Table 3: Across methods, sign consistency is high for concept shift (CS). Proposition 4.2 also identifies conditional informativeness (CI) as important for generalization, and XGB has the strongest value on this term. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

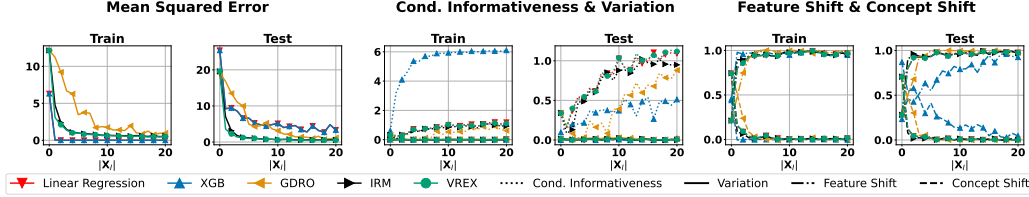


Figure 5: In the synthetic setting, adding more proxy variables $\mathbf{X}_I$ for U that are informative to Y reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

(see Appendix § B). We evaluate how the terms in the decomposition in (1) affect ID and OOD performance when going from one covariate subset to another covariate subset. To this end, we use what we call the *sign consistency metric*, which works as follows. When we move from one covariate set to a larger covariate set, for any term in (1) with a positive coefficient, such as conditional informativeness, we count how often its value rises when accuracy improves. For negatively weighted terms, such as concept shift, we count how often its value decreases when accuracy improves. The metric is the fraction of observations where a term’s change aligns with its *beneficial* direction, thus capturing how reliably it contributes to better generalization. See Appendix § B for the formal definition. Table 3 shows that both conditional informativeness and concept shift exhibit high sign consistency with accuracy gains. While reducing concept shift is crucial, XGB further enhances conditional informativeness compared to other methods, yielding additional performance improvements. Dataset-specific results are presented in Appendix § C.

Because real-world datasets rarely reveal whether *all available covariates* include *every relevant or sufficiently informative* variable, we also test the hypothesis in synthetic data with known causal structure: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ contains informative covariates caused by the hidden confounder U . In this controlled setting, adding $\mathbf{X}_I$ increases conditional informativeness and feature shift while reducing concept shift, as shown in Figure 5. The variation term remains zero because of the simplicity of the synthetic problem. Appendix § D gives the data-generation process and additional results.

6 Discussion, limitations, and future work

In summary, the predictive-information decomposition in Equation (1) identifies factors that can limit OOD generalization across environments. The analysis shows that hidden confounding shifts, visible through changes in covariate and outcome distributions, can entangle covariate, label, concept, and feature shifts (inequalities in (2), (3)). Under such conditions, treating each shift type in isolation is too narrow, and no finite set of observed covariates or proxies can completely remove ambiguity from unobserved confounders. We show that learning environment-specific input–output mappings can be more appropriate than enforcing a single invariant predictor (Equation (4)). These results support a broader view of OOD robustness in which invariance, informativeness, proxy availability, and confounder overlap jointly determine performance. Important directions include: (i) algorithms that remain effective under limited confounder overlap, (ii) methods for quantifying the cost–accuracy trade-off of acquiring informative covariates, (iii) approaches to entangled shifts that avoid untestable proxy assumptions, and (iv) models of ambiguity from unobserved confounders that can inform new OOD-robust paradigms.

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671 Appendix

672 A Proofs of propositions

673 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 674 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 675 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

676 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 677 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

678 *Proof.* The causal graph in Figure 3 implies a general redundancy property for the substructure
 679 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$: once either $\phi(\mathbf{X})$ or $\mathbf{X}$ is known, $\hat{Y}$ adds no further information about Y .
 680 The path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$ is blocked by either $\{\phi(\mathbf{X})\}$ or $\{\mathbf{X}\}$, giving $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and
 681 $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. We can therefore expand $I(Y; \hat{Y})$ as follows.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

682 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

683 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

684 The analogous decomposition for $\mathbf{X}$ follows by using $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introducing $\mathbf{X}$ rather than
 685 $\phi(\mathbf{X})$ in the same expansion. $\square$

686 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 687 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 688 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

689 *Proof.* Using the causal graph in Figure 3, the proposition is established through two causal directions
 690 that cover the relevant observed relationships: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

691 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 692 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

693 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 694 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 695 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

696 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

697 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

698 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

699 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 700 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 701 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

702 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 703 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

704 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

705 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

706 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

707 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 708 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

709 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 710 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

711 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
712 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
713 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
714 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

715 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
716 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
717 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
718 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
719 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

720 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

721 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

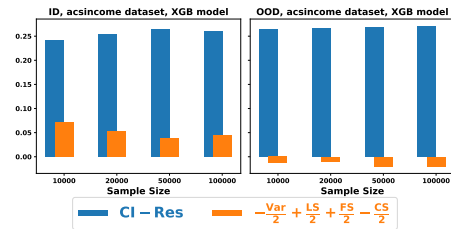
722 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
723 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
724 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
725 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
726 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

727 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
728 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
729 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
730 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
731 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

732 B Experimental setup

733 This section specifies the experimental setup used for all reported real-world benchmark results.
734 Random seeds have statistically insignificant effects in many of these settings [24]. For the results
735 in the main paper, we report the mean and standard deviation across five random hyperparameter
736 values (sampled from their respective domains in Table B1) for each dataset-method combination.
737 Since we evaluate models on ID test and OOD test data, to evaluate mutual information terms, we
738 use the nonparametric entropy estimation toolbox [37, 77, 78]. Following [24], we evaluate mutual
739 information using both in-distribution (ID) test data (from training domains) and out-of-distribution
740 (OOD) test data (from test domains).

741 For consistency across datasets and domains,
742 we select 20,000 random samples from each ID
743 and OOD test split when available, with smaller
744 datasets using all available samples. Across the
745 evaluated settings, sample size has only a lim-
746 ited effect on the estimated mutual-information
747 terms. For instance, as shown in Figure B1, the
748 terms in Equation (1) computed for the represen-
749 tations of the XGB model on the *income* dataset
750 stayed relatively consistent as we vary sample
751 size. Features $\phi(\mathbf{X})$ are extracted from the layer
752 preceding the classification head; for XGBoost,
753 we combine output margins and SHAP values from the model. All experiments were conducted on a
754 single NVIDIA RTX A6000 GPU.



755 Figure B1: Predictive-information terms remain
756 relatively stable across sample sizes on the *income*
757 dataset.

758 To study informative covariates, we use the covariate categories introduced by Nastl and Hardt [57]:
759 *causal*, *arguably causal*, and *all covariates*. Causal covariates have a well-supported directional
760 influence on the target and are unlikely to be affected by reverse causation. Relying only on this
761 conservative set can omit relevant parents when domain knowledge is incomplete. Arguably causal
762 covariates cover variables with less certain relationships, including variables that are known causal,
763 plausibly bidirectional, or likely but unconfirmed causes. These categories approximate causal

Table B1: Hyperparameter search spaces used across the evaluated model classes.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

761 parents using available knowledge, while allowing that relationships may remain confounded. Some
 762 datasets also include *anti-causal* covariates, where the target likely affects the covariate rather than
 763 the reverse. The *all covariates* set includes every observed variable regardless of causal status. We
 764 define the sign consistency metric from the main paper § 5 as follows. For each $(m, \sigma_m) \in M$ where
 765 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
 766 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

767 Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates
 768 and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and

test data. $\mu_s^{tr/te}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{causal, arguably\ causal, all\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section provides dataset-level real-world results that complement the aggregate findings in the main text. Figures C2, C3 compare *conditional informativeness – residual* with $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ for each dataset, whose aggregate appears in Figure 4 of the main paper. Consistent with Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall model performance. Under potential hidden confounding shift in real-world data, however, *conditional informativeness – residual* contributes more to predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. This supports a broader emphasis on conditional informativeness for OOD generalization when hidden confounding shift is plausible.

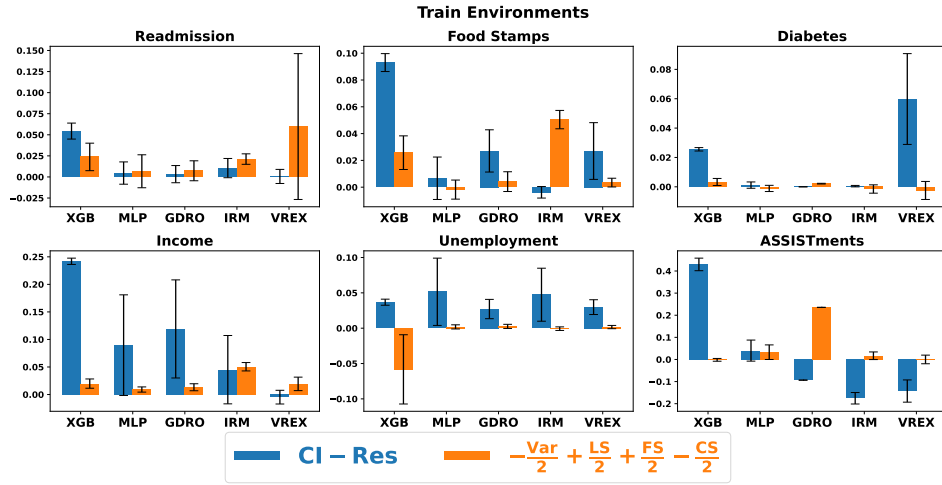


Figure C2: Dataset-specific decomposition results on ID data.

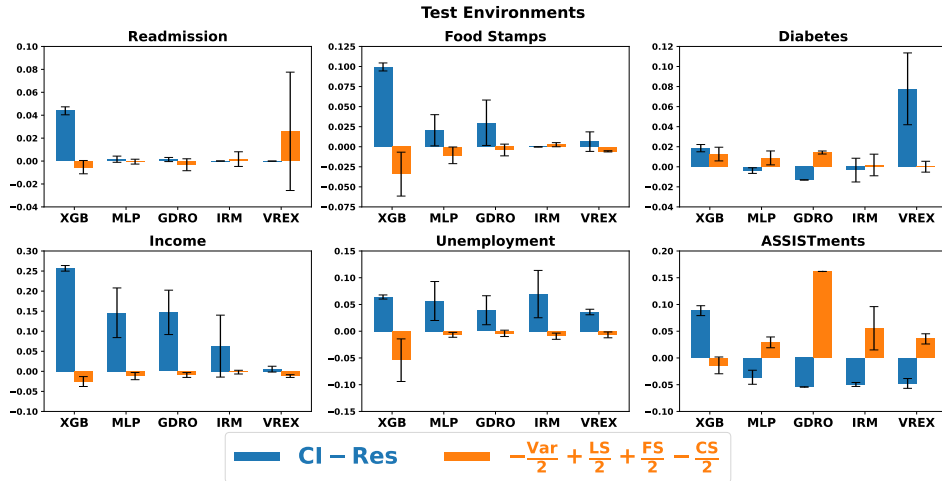


Figure C3: Dataset-specific decomposition results on OOD data.

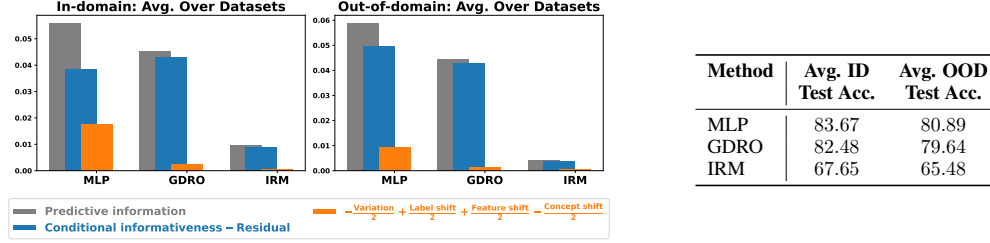


Figure C4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

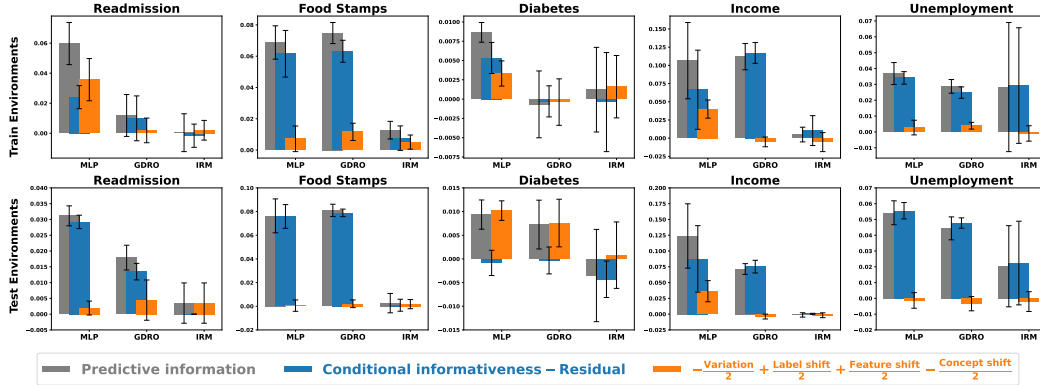


Figure C5: Dataset-specific comparison of predictive-information terms.

Table C2: ID and OOD test accuracies for the evaluated models and datasets.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further compare the terms in the information decomposition (1) for each method and dataset. The second-column values (*variation*) and third-column values (*feature shift*) are similar in magnitude, so their difference contributes little to the overall decomposition in Equation (1). Likewise, the fourth-column values (*label shift*) and fifth-column values (*concept shift*) are similar in magnitude and nearly cancel in Equation 1. Across nearly all settings, *conditional informativeness* - *residual* is the dominant contributor to predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of *blue* and *orange* terms is

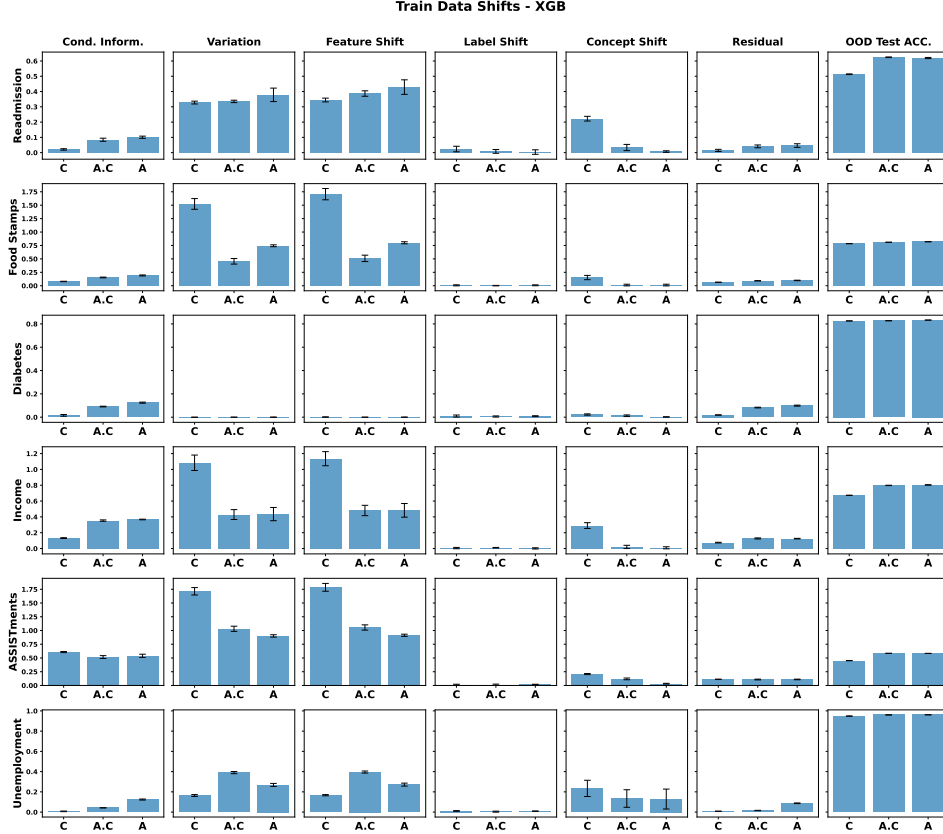


Figure C6: Train-data predictive-information decomposition for XGBoost across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

796 positively correlated with the overall model performance. At the same time, due to potential hidden
797 confounding shift in real-world data, the contribution of the difference *conditional informativeness* –
798 *residual* is higher towards the overall predictive information when compared to the contribution of
799 $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. Dataset-specific results shown
800 in Figure C5 and Table C2 also show similar trends.

801 **Potential limitations with predictive information decomposition analysis:** While predictive
802 information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, it does not
803 guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label
804 (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to perfect
805 anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction*
806 *in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional
807 informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift
808 $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition
809 remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it
810 reveals whether predictive information stems from generalizable patterns or environment-specific
811 artifacts, enabling targeted improvements while maintaining interpretability.

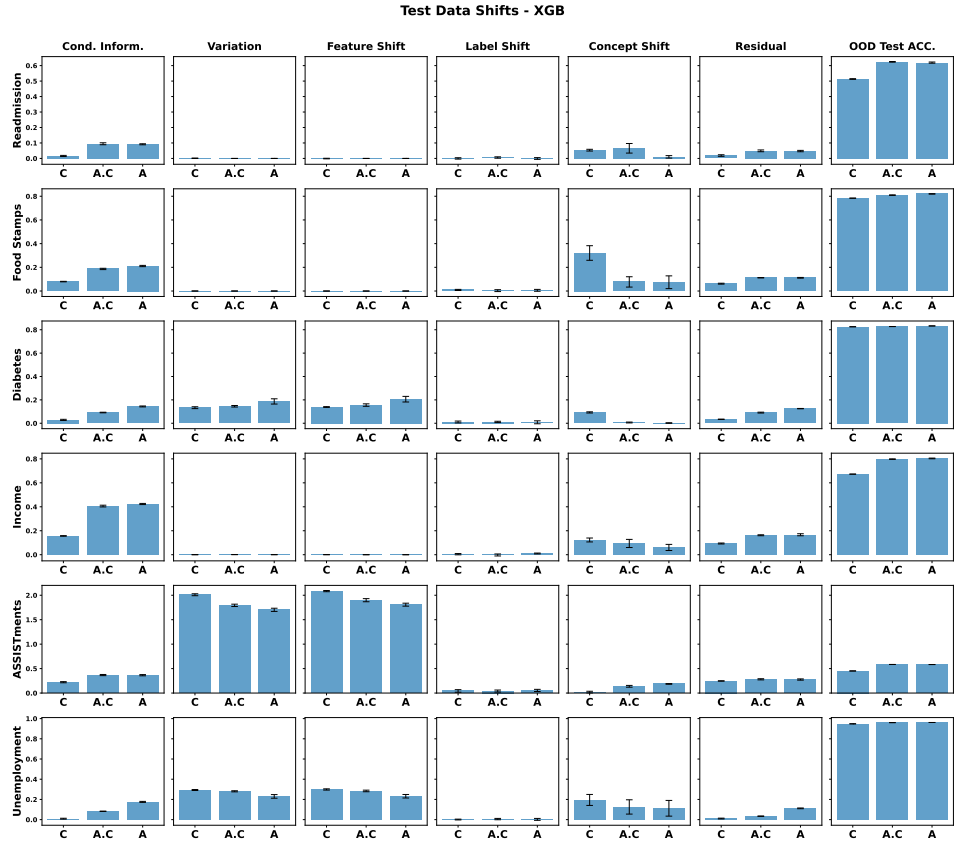


Figure C7: Test-data predictive-information decomposition for XGBoost across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

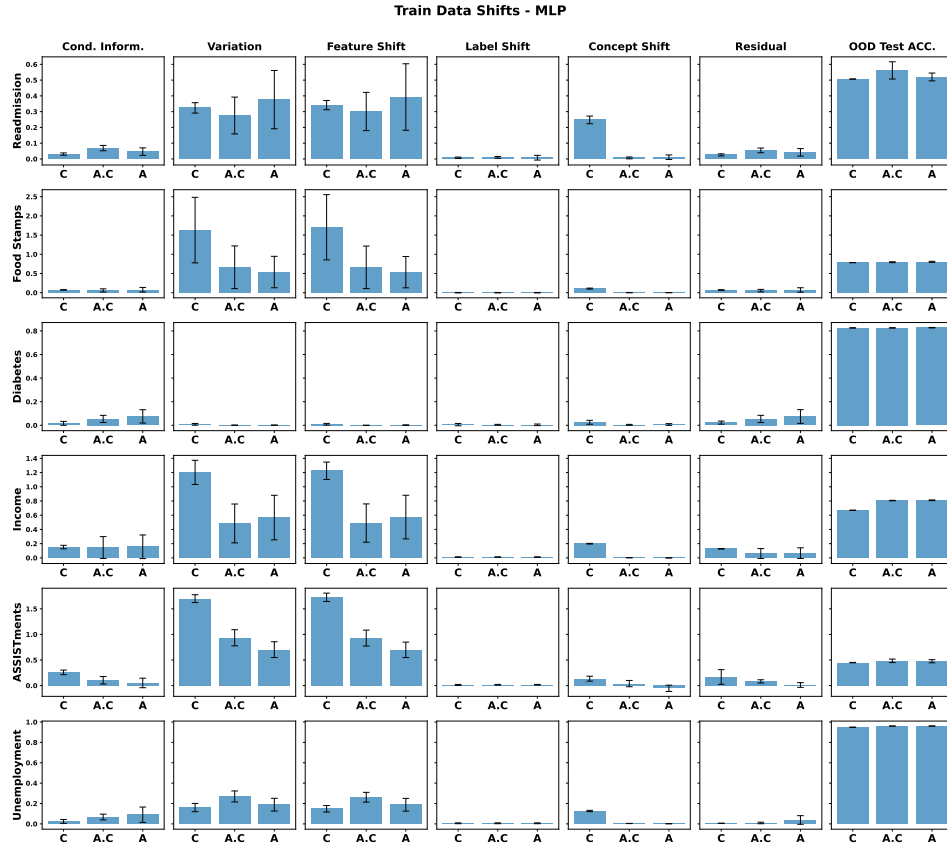


Figure C8: Train-data predictive-information decomposition for MLP across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

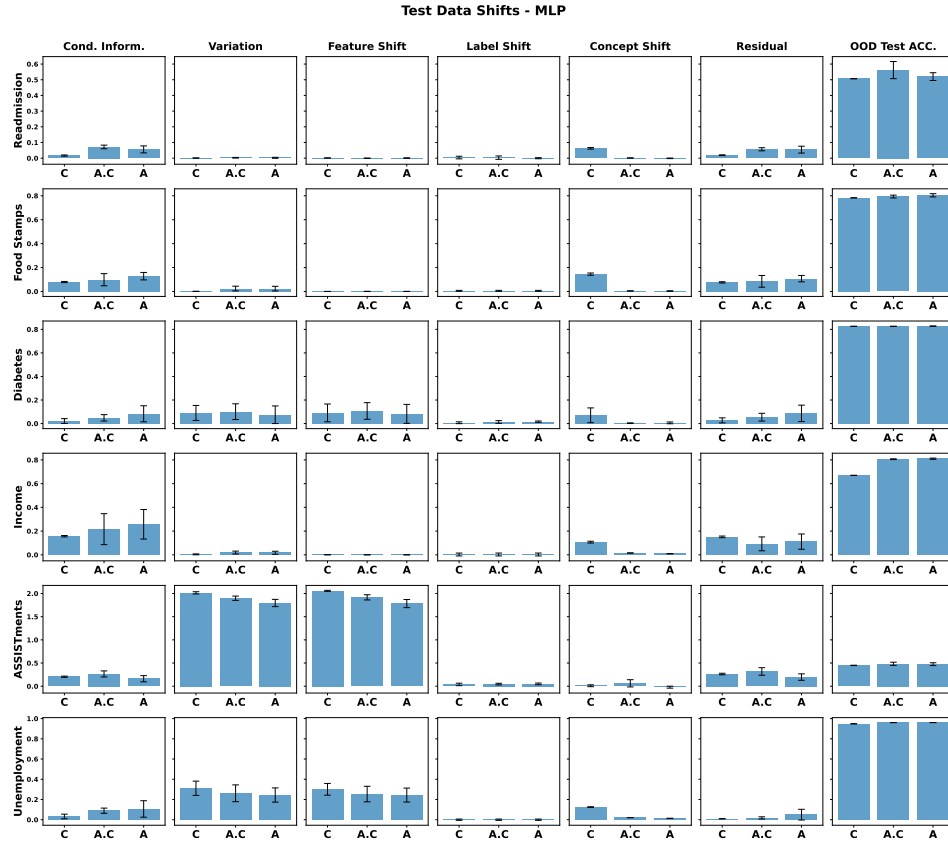


Figure C9: Test-data predictive-information decomposition for MLP across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

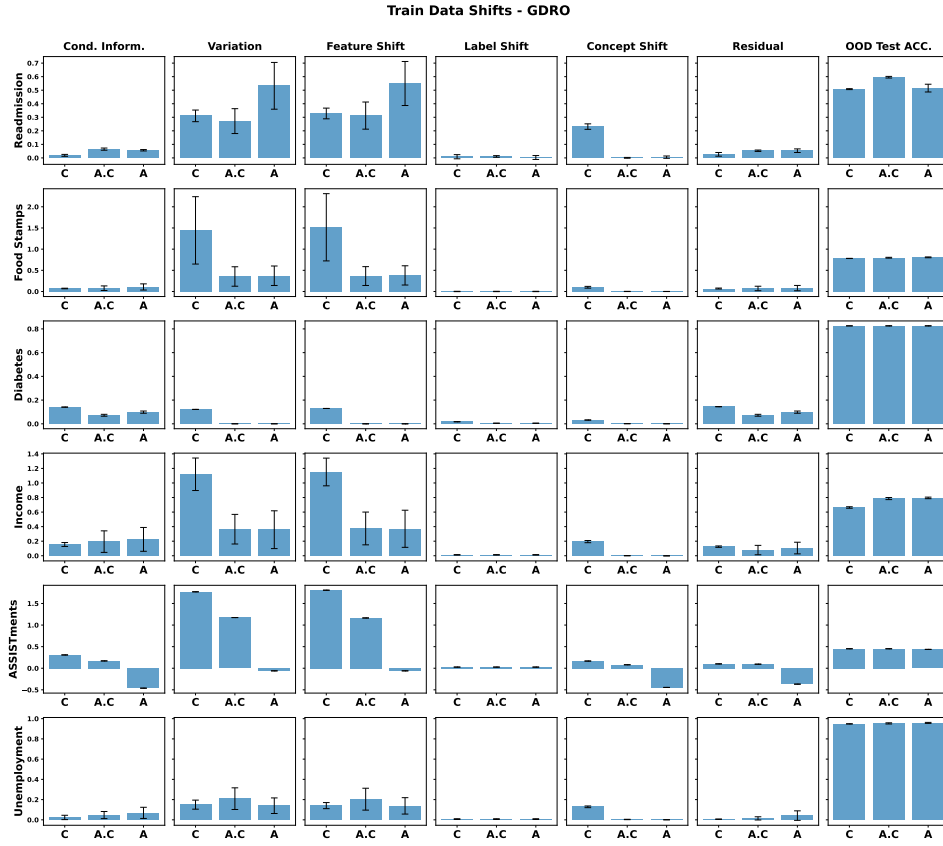


Figure C10: Train-data predictive-information decomposition for GDRO across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

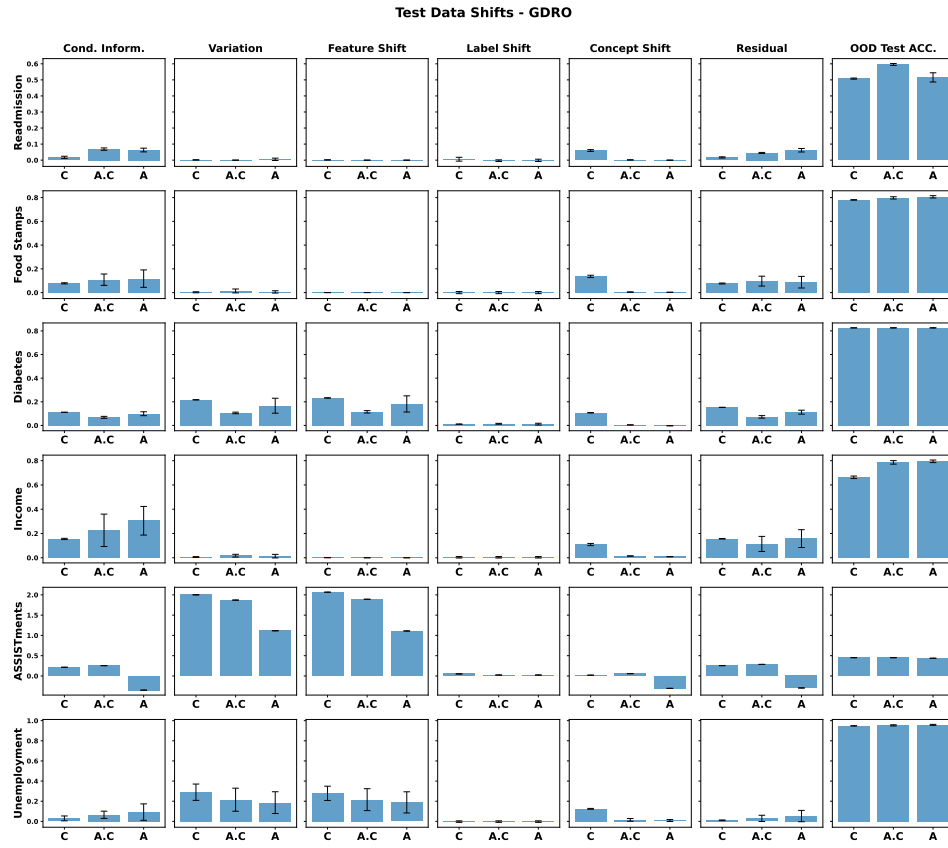


Figure C11: Test-data predictive-information decomposition for GDRO across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

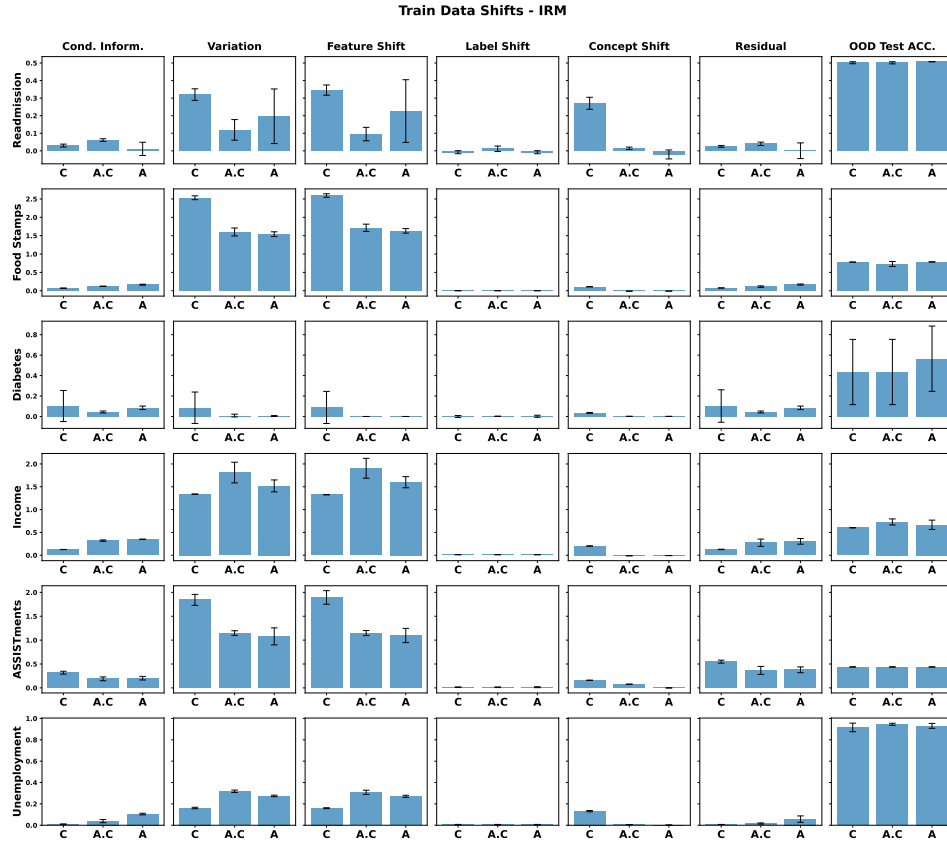


Figure C12: Train-data predictive-information decomposition for IRM across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

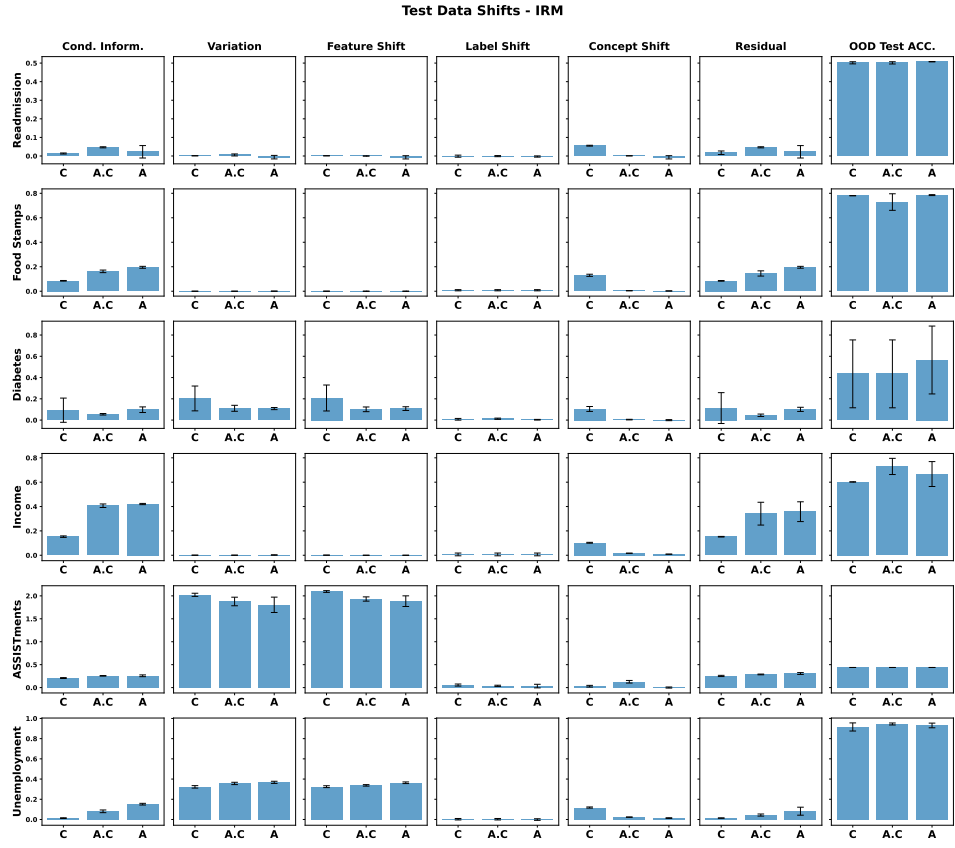


Figure C13: Test-data predictive-information decomposition for IRM across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

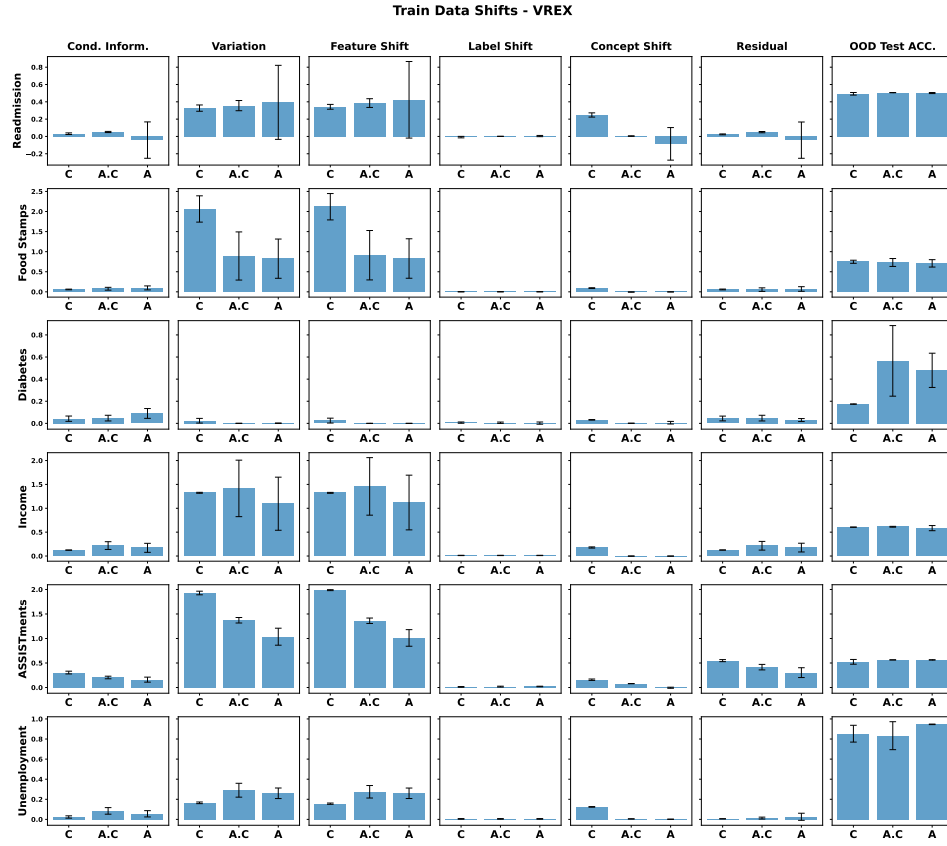


Figure C14: Train-data predictive-information decomposition for VREX across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

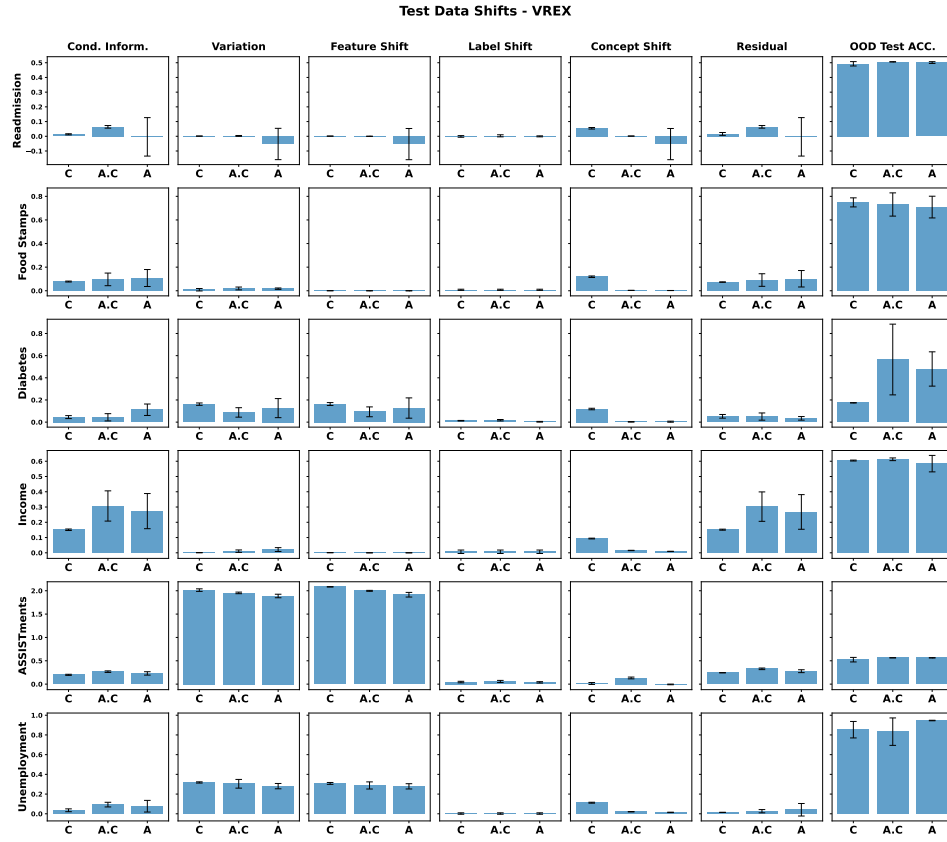


Figure C15: Test-data predictive-information decomposition for VREX across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

D Results on synthetic datasets

We use synthetic datasets to isolate three mechanisms that are difficult to verify in real-world data: (i) the effect of different informativeness criteria, (ii) the role of confounder overlap across environments, and (iii) the influence of the causal-to-anti-causal variable ratio on performance. The data-generating process follows the structural equations in (12), with the causal structure $U \rightarrow X$, $U \rightarrow Y$, $U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is an unobserved confounder, X is an observed covariate, Y is the target variable, and $\mathbf{X}_I$ represents additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

We evaluate several functional forms for f_X , f_i , and f_Y and report the corresponding results. Domain shifts are induced by varying the environment-specific confounder mean μ_u^e , which lets us study how increasing distributional separation affects performance across environments.

Informativeness vs. accuracy: In this set of experiments, we begin with linear structural equations: $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$. For these experiments $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test environments. For the results presented in Figure 5 of the main paper, we consider one hidden confounding variable with $|\mathbf{X}_I| = 20$ informative covariates. As we increase the number of informative covariates from 0 to 20, we observe: (i) a reduction in mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while (iv) decreased concept shift. We extend these findings by examining two additional informativeness criteria. First, we analyze the case with one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$) across varying noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$. Figure D16 demonstrates that as noise decreases from 2.0 to 0.0, we observe: (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, alongside (iv) decreased concept shift. Consistent with other synthetic experiments, the variation term remains zero throughout. Finally, we investigate a setting with 20 hidden confounders and corresponding 20 informative covariates, using modified structural equations $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17 shows that as we introduce informative covariates corresponding to hidden confounders, we again observe: (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while (iv) decreased concept shift. These experiments demonstrate how different informativeness criteria help in achieving OOD generalization.

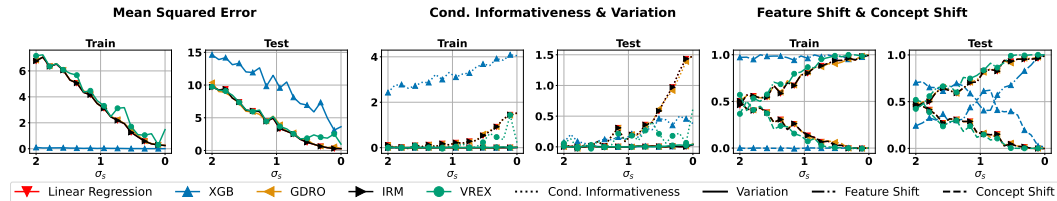


Figure D16: Adding a low-noise proxy variable reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift in the synthetic setting.

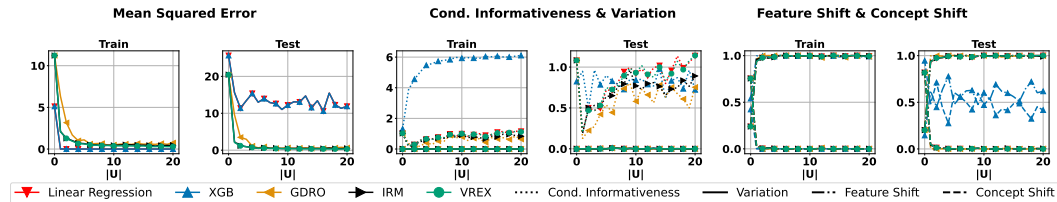


Figure D17: Adding more proxy variables $\mathbf{X}_I$ for hidden confounders $\mathbf{U}$ that are informative to Y reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

Overlapping confounder support vs. accuracy: As established in the main paper, recent work by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that

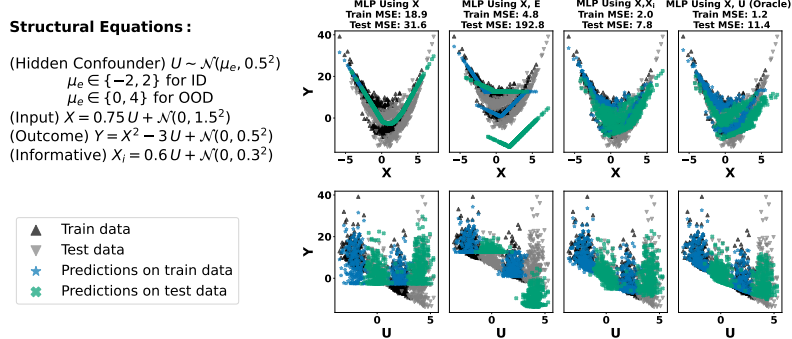


Figure D18: MLP performance in the low-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

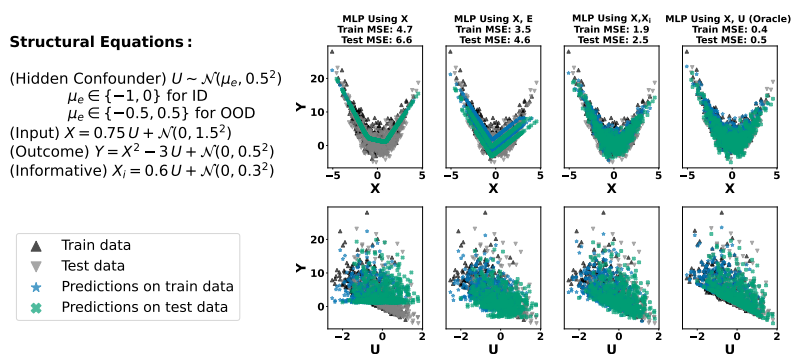


Figure D19: MLP performance in the high-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

assumes test confounder support remains within training support. While our results demonstrated successful learning of correct relationships in synthetic linear data when sufficient information about hidden confounders exists in observed covariates, we now extend this analysis to nonlinear data using MLP and XGB models under both low and high confounding support conditions (Figures D18–D21). In low confounding overlap settings (characterized by distant μ_e between ID and OOD data), models relying solely on X learn an inadequate global function that fails to distinguish environments. Performance improves when incorporating environment-specific summary statistics of observed covariates, which helps capture environment-specific relationships, but the most significant gains occur only when leveraging additional informative covariates, underscoring their importance. For completeness, we include both oracle model results and U - Y relationship scatter plots to elucidate the learned input-output mappings. These experiments confirm better performance under high confounding overlap as expected, but more importantly, reveal promising results in the low-overlap regime - a previously unstudied scenario that challenges the common support assumption in the literature.

Number of causal vs anti-causal variables: Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) in real-world datasets, coinciding with the prevalent causal structure $\mathbf{X} \rightarrow Y$ in tabular data. To test if baseline superiority stems from the causal structure: $\mathbf{X} \rightarrow Y$, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). Results (Figure D22) show baseline resilience even under the causal structures $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, though no method dominates universally (Figures D23, D24). Baselines achieve superior PMA-OOD scores [24] (fraction of maximum OOD accuracy). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43],

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

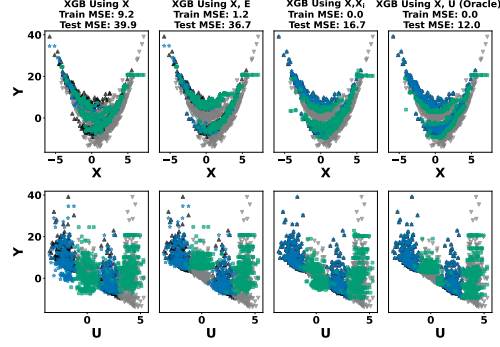


Figure D20: XGB performance in the low-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

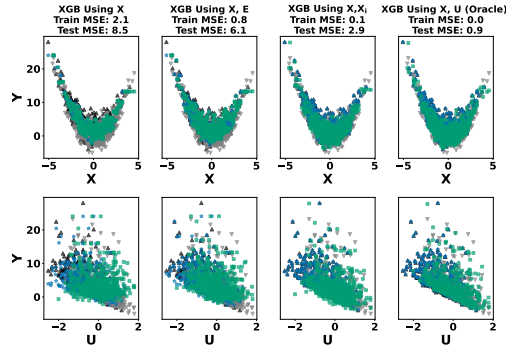


Figure D21: XGB performance in the low-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

867 DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods
868 (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_s|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

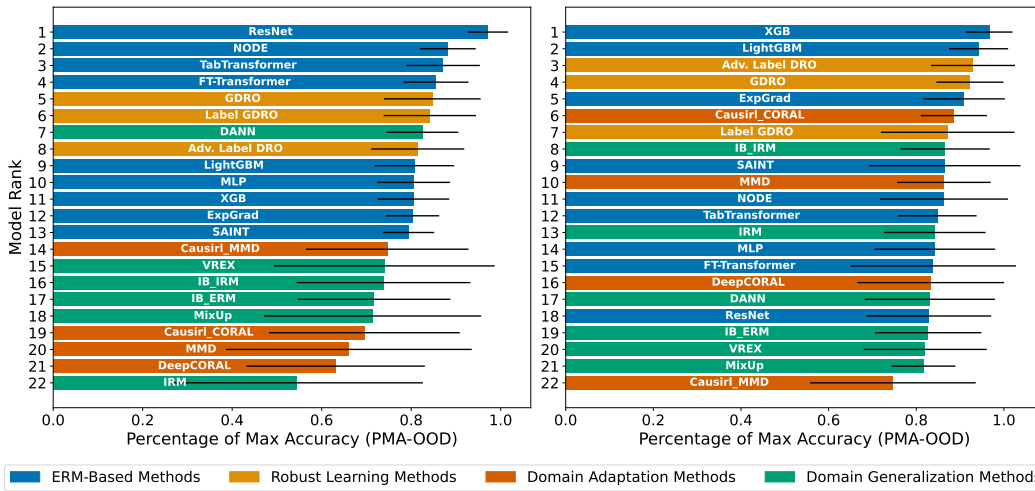


Figure D22: ERM-based methods achieve higher percentage-of-maximum OOD (PMA-OOD) test accuracy on these synthetic datasets, consistent with the real-world benchmark pattern reported by [24].

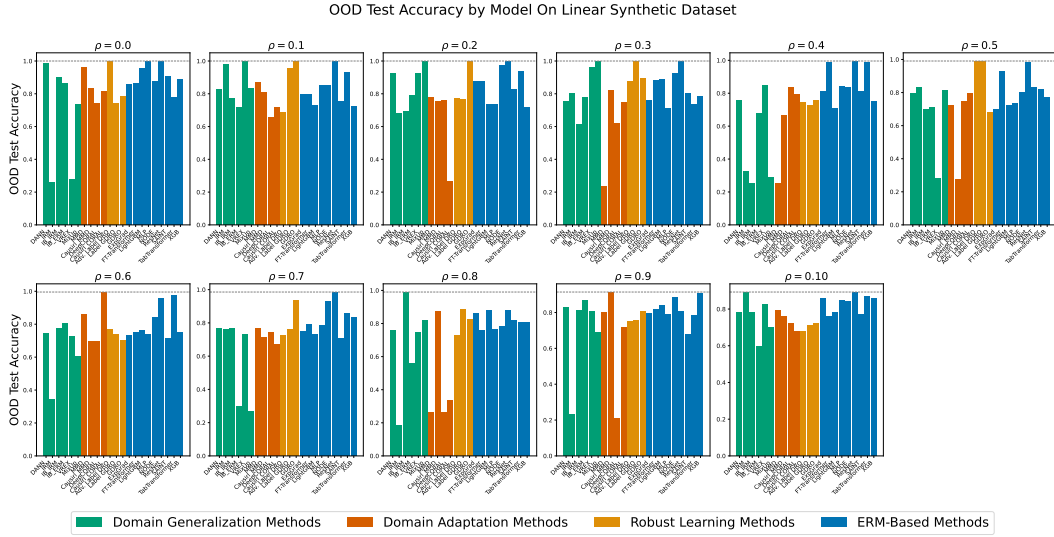


Figure D23: OOD test accuracy in the linear setting for each model as ρ varies.

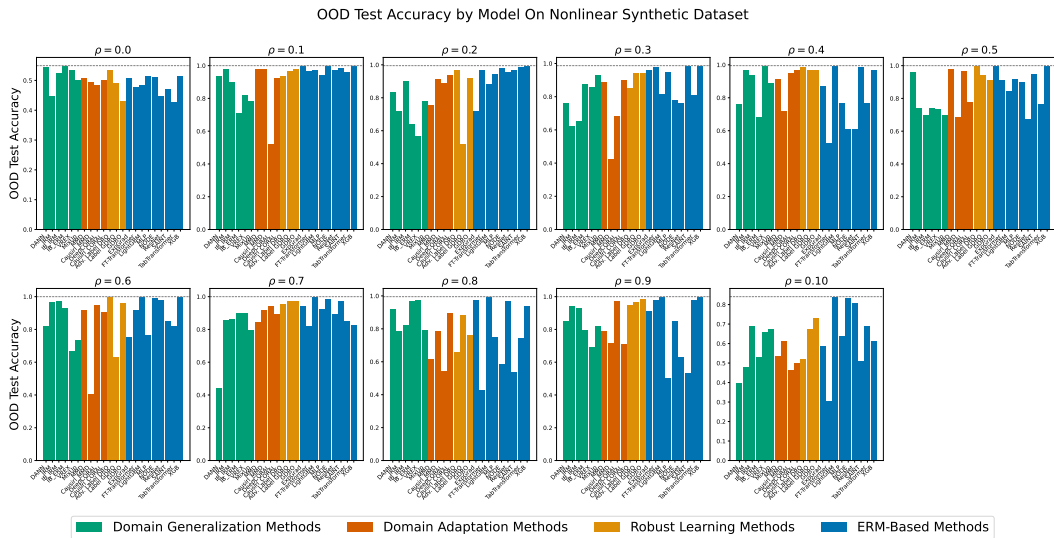


Figure D24: OOD test accuracy in the non-linear setting for each model as ρ varies.

869 E Qualitative analysis of hidden confounding shift in real-world datasets

870 This section qualitatively examines hidden confounding shift in real-world datasets. Tables E3-E18
 871 list plausible hidden confounders that may influence observed covariates across several datasets [57];
 872 dataset details are available in [24, 57]. We use GPT4o [1] to generate these candidate confounders,
 873 so the lists should be read as illustrative rather than exhaustive. The exercise supports the broader
 874 argument that hidden confounding shift can plausibly arise across domains represented in real-world
 875 tabular benchmarks.

876 **Prompt:** We use the following prompt to elicit possible hidden confounders from GPT4o.

877 For a target variable, you will be given lists of causal, arguably
 878 causal, anti-causal, and spurious covariates. Also, you will be provided
 879 with environment variables such that different datasets are collected in
 880 environments induced by these environment variables. Provide possible
 881 hidden confounders that cause any of the target, causal, arguably causal,
 882 spurious, and anti-causal covariates. That is, the hidden confounders
 883 should influence the distribution of their children in different domains.
 884 Ensure hidden confounders are not a part of the given variables, and your
 885 explanation should include how hidden confounders influence their children
 886 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.